Does your money go through the roof?

Insulation – does it stop heat transfer and save money? You probably know the answer to this question but you need to have figures to back up your answer.

As a class we will test the effectiveness of several insulating materials:
These may include: Fibreglass, cotton wool, newspaper, shiny bubble wrap

Write down the material you will use
_____

You will need: a small metal can with a lid
A polystyrene holder
A thermometer
A stopwatch
Access to hot water (100ml)

Method:

- Firstly work out what variables are to be kept the same. Write them in the space below

- Work out how a control experiment can be performed. Write your answer below

- Boil the kettle
- Place your test materials around the can and place this inside the polystyrene holder
- Pour 100ml of water into your can

- Place the lid on the can and push a thermometer through the hole
- Record the temperature of the water, this is time ‘0’
- Record the temperature of the water every 2 mins in the table overleaf,for up to half an hour

Results :

Time (mins)	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
Temperature Test can (°C)																
Temperature Control (°C)																

The results of the activity are more obvious if you draw a graph. Use graph paper and draw a line graph. (Remember: the variable you changed goes on the bottom, the one you measured goes on the vertical axis)

Conclusion: Does your test material decrease the transfer of heat and save you money?

Questions:

How does your test material compare with that of other groups?

Why did the cans have to be the same size?

Why is it useful to draw graphs of the data you obtained?

How can ceiling insulation save people money?